**Code No: 117AN** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year I Semester Examinations, November/December - 2018**
**ARTIFICIAL INTELLIGENCE**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

**PART- A**

**(25 Marks)**

1.a) What are the functionalities of an agent system? [2]
b) How can we avoid ridge and plateau in hill climbing? [3]
c) What are limitations in using Propositional Logic to represent the knowledge base? [2]
d) Represent the following sentence in Predicate form: "All the children like sweet". [3]
e) Compare Expert system with Traditional system. [2]
f) State and define conditional probability. [3]
g) Define supervised learning and unsupervised learning. [2]
h) Differentiate ANN with Biological NN. [3]
i) What is semantic web? [2]
j) What is the importance of Parser? [3]

**PART-B**

**(50 Marks)**

2.a) What are sub-areas of AI?
b) Explain a simple approach to play Tic-Tac-Toe problem. [5+5]

**OR**

3. Discuss about different Heuristic Search Techniques. [10]

4. What is Resolution? Explain resolution in propositional logic with illustration. [10]

**OR**

5.a) Differentiate between Procedural and Declarative Knowledge.
b) Describe about Forward Chaining System. [5+5]

6. Explain Rule based system Architecture with neat diagram. [10]

**OR**

7. Describe the following:
a) Dempster-Shafer Theory b) Certainty Factor Theory. [5+5]

8. Explain Inductive Learning and Deductive Learning with example. [10]

**OR**

9.a) Explain design issues of Artificial Neural Networks.
b) Discuss about Recurrent Networks with example. [5+5]

10.a) What is Parsing? Derive a parse tree for the sentence "Bill loves the frog".
b) Write about semantic analysis in NLP. [5+5]

**OR**

11. Explain in detail about several components of the Natural Language Processing. [10]

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